

Statistical and Machine Learning Perspectives on Exoplanets Prediction and Classification

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Abstract—Exoplanets are the planets orbiting stars out of our solar system. The exploration of such planets has become a growing field in astronomy but the analysis of the data related to such planets involves many challenges when traditional methods are used as this data is often vast and is obtained with the help of telescopes and space missions. This study proposes a methodology that aims to classify exoplanets, taken from datasets from the Kepler Objects of Interest (KOI) database and the NASA Exoplanet archive, as confirmed, candidate, or false positives. This approach makes use of modern analytical techniques, such as statistical analysis, probability theory, hypothesis testing, and machine learning (ML) algorithms, to facilitate the detection and classification of exoplanets. Clustering methods such as BIRCH and K-Means are also performed on the Planetary data. This study contributes to improving our knowledge and understanding of exoplanetary systems, thereby highlighting the relevance of modern analytical tools in the field of astronomy.

Index Terms—BIRCH, Clustering, Exoplanets, Hypothesis Testing, K-Means, KOI, Machine Learning Algorithms, NASA Exoplanet Archive, Planetary Systems, Probabilistic Models, Statistical Analysis

I. INTRODUCTION

Exoplanetary science or the study of distant worlds is being propelled forward by rapid strides in the field of astronomy. However, understanding planetary diversity, formation, and potential for life beyond Earth is an interesting field of study although it becomes a boring task using traditional approaches. The technical difficulties involved in obtaining clear images make direct imaging a difficult technique [1] [2].

The confirmed exoplanets can be clustered to facilitate easy comprehension through k-means and BIRCH clustering techniques [3] [4]. The KOI database [4] and the NASA Exoplanet archive provide reliable data on celestial objects. Modern statistical techniques such as probability theory, univariate and bivariate analysis as well as dimensionality reduction techniques like Principal Component Analysis (PCA) assist in analyzing and accurately classifying exoplanets into confirmed, candidates and false positives. To optimize classification, machine learning algorithms such as random forests, decision trees, hidden Markov models, and Gaussian naive Bayes can be used.

The present system intends to make good use of these latest technological tools to gain better insight into the characteristics of exoplanets.

II. LITERATURE SURVEY

This literature survey examines the use of advanced computational techniques across various fields, including exoplanet detection, software engineering, communication networks, and natural language processing.

Machine learning (ML) techniques are improving exoplanet prediction accuracy and efficiency. Cuéllar *et al.* [1] and Ghadekar *et al.* [2] use various ML approaches, including logistic regression, random forests, CNNs, and Siamese architectures, to achieve high accuracy rates despite constraints on datasets. Bhamare *et al.* [4] extended this research by obtaining 99% prediction accuracy with the random forest technique, despite measuring astronomic and radial velocity signals owing to stellar activity.

Jin *et al.* [3] present a deep-learning technique for detecting exoplanets, which improves precision and recall for both real and synthetic data, despite restricted dataset availability. Karim *et al.* [5] highlights RNNs and deep learning architectures, such as ASTRONET in analyzing exoplanet habitability by boosting identification accuracy while stressing need for more research.

Jara-Maldonado *et al.* [6] provide a comprehensive framework for testing ML models in exoplanet discovery through transit signals, improving execution time and identification performance. Srivatsha and Assaf [7] employ ML models to classify Kepler Objects of Interest (KOIs) as exoplanets or false positives, obtaining high accuracy and identifying key properties. Matchev *et al.* [8] and Winter *et al.* [9] investigated unsupervised ML algorithms for understanding exoplanetary system structures, focusing on classification methods, anomaly detection, and the impact of star clustering. Mousavi-Sadr *et al.* [10] used the Titius-Bode relation and a Markov Chain Monte Carlo technique to predict exoplanets in multi-planetary systems.

Priyadarshini *et al.* [11] and Malik *et al.* [12] suggest a CNN-based ensemble model employing spectral diversity to improve performance, as well as an SVM and CNN model for exoplanet discovery that addresses data augmentation. Gilbert *et al.* [13] proposed an information-theoretic framework to categorize exoplanet system structures into four kinds based on formation paths and relationships with star metallicity, laying

the groundwork for additional research. Bryson *et al.* [14] suggested a probabilistic approach to address the completeness and reliability in exoplanet catalogs, concentrating on star properties in the occurrence rate of predictions.

Mathur *et al.* [15] investigated deep learning algorithms for exoplanet identification and habitability prediction. They found significant classification accuracy gains using data augmentation and Siamese architecture. Tang *et al.* [16] used multi-satellite data to analyze exoplanets and their host stars, identifying trends in size distributions and connections with stellar metallicity. Their findings emphasize the importance of machine learning in astronomy and knowing wide-orbit planets and star multiplicity in furthering planetary formation and habitability research.

Hara *et al.* [17] and Jones *et al.* [18] use statistical techniques and Gaussian process models to discover exoplanets based on radial velocity. They handle issues such as noise impact and stellar activity. Herur *et al.* [19], Hayes *et al.* [20], Pradeep Kumar *et al.* [21], Balaji *et al.* [22], Sanjanasri *et al.* [23], and Surendran *et al.* [24] all contribute to the exoplanet prediction research landscape. They cover topics including hybrid KNN models, unsupervised machine learning categorization, wideband antenna technology, spectral data pre-treatment, machine translation augmentation, and edge-centric predictive modeling. These studies enhance the field of exoplanet prediction by utilizing diverse machine learning and statistical methodologies, addressing problems and offering up new options for future research and growth.

The literature highlights the transformative impact of machine learning and deep learning across disciplines, noting progress but also challenges like data scarcity, model interpretability, and scalability.

The major contributions of this paper are :

- The thorough analysis of the different features of the KOI dataset.
- Implementation of Probabilistic and Statistical Models for feature selection and analysis.
- Feature selection and implementation of different methods of clustering on the Planetary Systems data.

III. METHODOLOGY

A. Proposed Methodology

The project aims to predict exoplanets using the KOI dataset [25] and cluster confirmed exoplanets with the Planetary Systems dataset [26]. Both datasets are preprocessed for missing values, outliers, and normalization. Feature analysis includes correlation, PCA, and univariate/bivariate analyses to refine feature selection.

Machine-learning models (Random Forest, Decision Trees, AdaBoost) are trained and evaluated on both all features and selected features to assess the impact of feature selection on accuracy. Probabilistic models (SVM, HMM, Gaussian Processes, Naive Bayes) are also tested.

Statistical techniques (Chi-square, ANOVA, Fisher Exact Test) assess feature significance, while clustering algorithms (K-means, BIRCH) are applied to the Planetary Systems

dataset to identify trends and patterns among confirmed exoplanets.

B. Data Preprocessing

The KOI dataset [25], initially with 9,564 rows, was preprocessed by removing redundant columns and irrelevant attributes, replacing missing values with column averages, and scaling with `StandardScaler()`. The data was split 70:30 for training and testing. The Planetary Systems dataset [26], initially about 34,000 rows, had 10% removed due to high missing values and columns with over 75% blanks. Remaining missing values were filled with means or modes, categorical columns were labeled, and data was scaled to prepare it for analysis.

C. Feature Selection and Analysis on the KOI Data

1) *Correlation Analysis*: Correlation analysis examines the correlations between variables [4] [3]. In this study, correlation analysis was used to investigate the correlations between the 19 features in the KOI dataset's training data. This included calculating covariance for all features and showing the associations using a heatmap.

2) *PCA (Principal Component Analysis)*: Principle Component Analysis (PCA) reduces dimensionality by transforming original features into new, independent components called principle components, which capture the most important variation in the dataset [4]. In this study, PCA was used to identify essential variables in the KOI dataset while keeping a pre-determined number of principle components. The cumulative explained variance ratio was then shown to show how each component contributed to the overall variance.

3) *Univariate Analysis*: Univariate analysis is a fundamental type of data analysis that focuses on a single variable [3]. It gives information on the distribution, central tendency, and spread of the data by focusing on one variable at a time. This form of analysis is critical for feature analysis and selection because it identifies features with large variability that may be useful for predictive modeling while avoiding the complexities of multivariable interactions.

4) *Bivariate Analysis*: Bivariate analysis explores the relationship between two variables to uncover correlations, patterns, and interactions that impact the target variable. In astronomy, 'ra' (right ascension) and 'dec' (declination) are essential for locating celestial objects but may not influence exoplanet classification, as they provide positional rather than property-related information [3].

D. Machine Learning Models

Random forests resist overfitting and offer accurate results across diverse datasets, while decision trees enhance interpretability through a tree-like structure. AdaBoost combines weak classifiers into a strong classifier. Performance is measured by accuracy, precision, recall, and F1 score, which collectively assess the quality of predictions.

The study evaluates these algorithms on both the full set of 19 features and a subset of 11 key features, aiming to see how

feature selection impacts performance. Evaluation metrics will help determine the best method and feature set for classifying exoplanets as candidates, confirmed, or false positives.

E. Probabilistic Methods on the KOI Data

Hidden Markov Models (HMM), Gaussian Processes, and Logistic Regression are evaluated with both all features and a subset of 11 features to assess their impact on exoplanet classification.

Model performance is measured by accuracy (overall correct predictions), precision (true positives vs. all positives), recall (true positives vs. actual positives), and the F1 score (harmonic mean of precision and recall).

F. Statistical Analysis of KOI Data

1) *Chi-Square Test*: The chi-square test of independence determines if there is a significant relationship between categorical variables in a dataset. Applied to the 'koi_disposition' target variable, it assesses whether the variables are independent (null hypothesis) or significantly related (alternative hypothesis).

Key metrics in this test include the chi-square statistic, which measures the difference between observed and expected counts in a contingency table. A higher chi-square value indicates a stronger relationship between the variables. Degrees of freedom (dof) represent the number of independent pieces of information and are calculated based on the dimensions of the contingency table, affecting the chi-square distribution.

The p-value indicates the probability of obtaining a chi-square statistic as extreme as, or more extreme than, the observed value if the variables were independent. A low p-value provides evidence to reject the null hypothesis, suggesting a meaningful relationship between the variables. Expected frequency tables, which show counts assuming independence, are compared to observed frequencies to identify the most important predictor factors for exoplanet categorization.

2) *ANOVA - Analysis of Variance*: The Analysis of Variance (ANOVA) is a statistical tool for determining whether there are significant differences in the means of three or more groups. When applied to the dataset with the goal variable 'koi_disposition,' ANOVA analyses if various predictor variables have statistically significant differences in means across the categories of 'koi_disposition'.

The null hypothesis (H0) asserts that all groups means are equal, while the alternative hypothesis (H1) states that at least one group mean is different. The F-statistic compares the variance between group means to the variance within the groups. The p-value reflects the probability of observing these results if H0 is true.

3) *Fisher Exact Test*: Fisher's exact test is a statistical significance test that determines if two categorical variables have a nonrandom connection. This test is applied to the dataset with the goal variable 'koi_disposition,' and examines whether each feature has a significant correlation with the target variable.

Fisher's exact test calculates the odds ratio and p-value to assess the association between a feature and a target variable. The null hypothesis (H0) states there is no association while the alternative hypothesis (H1) suggests a significant relationship. A p-value below the significance level indicates rejecting H0, showing a significant association. A p-value above the significance level means H0 is not rejected, indicating no significant association.

G. Regression Models on KOI Data

Linear regression and logistic regression are two fundamental regression models used in statistical analysis. Linear regression predicts continuous outcomes by assuming a linear relationship between independent variables and the target variable. Logistic regression, on the other hand, is employed for classification tasks, modeling the probability of the target variable belonging to specific classes.

Evaluation metrics such as mean squared error (MSE), root mean squared error (RMSE), and R-squared score (R2) assess the performance of linear regression, providing insights into accuracy and model fit. Logistic regression, on the other hand, utilizes metrics like accuracy, precision, recall, F1 score, confusion matrix, and classification report to evaluate classification performance.

H. Clustering on Planetary Systems Data

K-means partitions data into clusters based on Euclidean distance. BIRCH uses a tree structure for hierarchical clustering, maintaining summary statistics for each cluster. PCA is used to reduce dimensionality, enhancing clustering by focusing on key features.

Evaluation metrics include the Davis-Bouldin Index (DBI) and Silhouette score. DBI measures cluster similarity, which lower values indicating better clustering. The Silhouette score ranges from -1 to 1, which higher scores indicating better defined clusters. Feature selection identified the top 6 most informative features for clustering. K-means and BIRCH were applied to these features, with and without PCA, to improve clustering accuracy and interpretability.

IV. RESULTS AND ANALYSIS

A. Correlation Analysis

Based on the heatmap and correlation matrix, specific features were identified for further analysis due to their significant correlations with each other. These features are 'koi_fpflag_ss', 'koi_fpflag_co', 'koi_fpflag_ec', 'koi_time0bk', 'koi_impact', 'koi_depth', 'koi_prad', 'koi_teq', 'koi_insol', 'koi_model_snr', 'koi_slogg', 'koi_srad', and 'koi_kepmag'. These features were selected because their interrelationships indicated that they might contain valuable information for subsequent modeling and analysis, helping to reduce redundancy and focus on the most informative attributes.

Fig. 1 shows the correlation matrix for all the features.

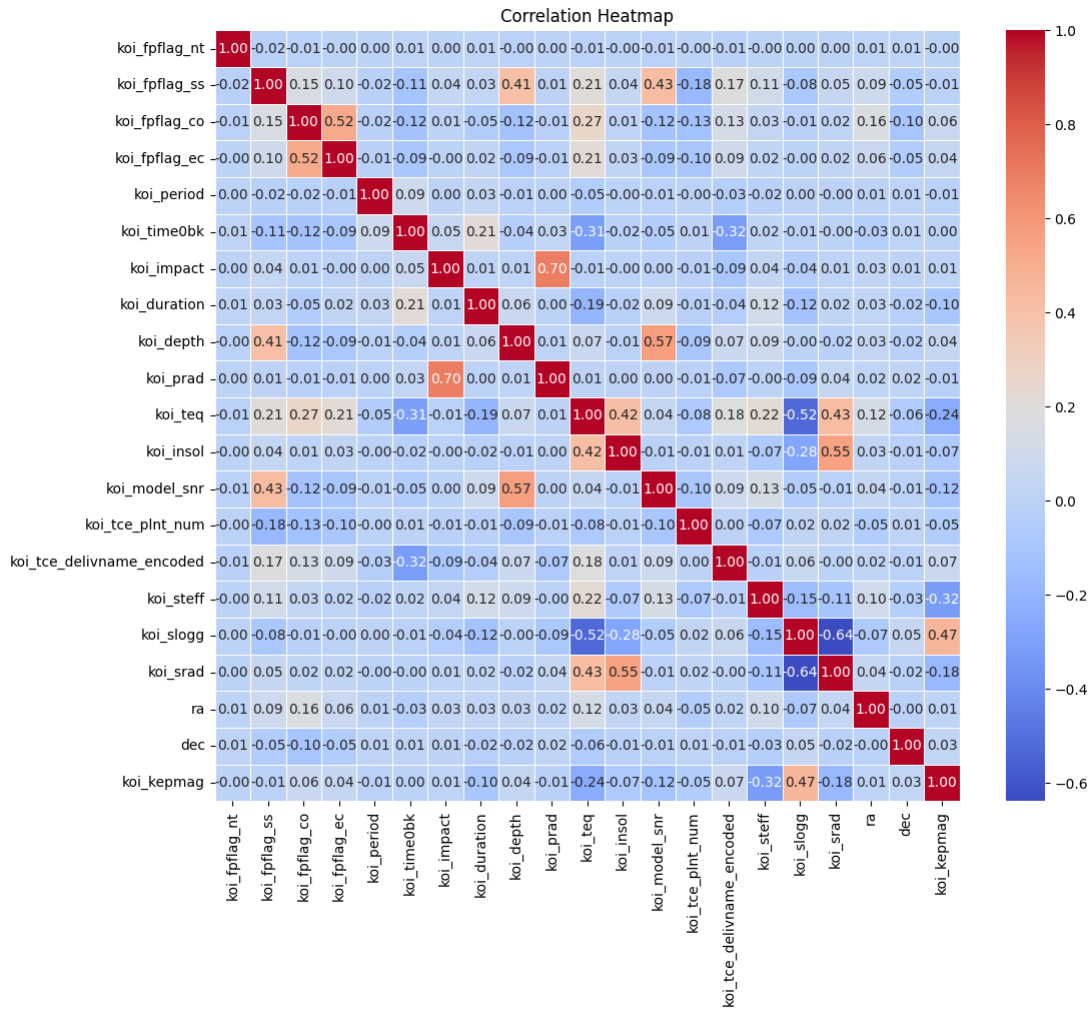


Fig. 1: Correlation Matrix

B. PCA Analysis

The plot shows the cumulative explained variance ratio by the number of principal components from PCA. The ratio rises quickly, nearing one, indicating the few components explain most of the dataset's variance. The curve then stabilizes, suggesting additional components offer minimal variance explanation which highlight's PCA's effectiveness in dimensionality reduction.

Feature importance analysis was conducted using a Random Forest classifier to further validate the importance of the features identified by PCA. The top features determined through PCA included 'koi_fpflag_nt', 'koi_fpflag_co', 'koi_model_snr', 'koi_fpflag_ss', 'koi_prad', 'koi_depth', 'koi_fpflag_ec', 'koi_impact', 'koi_period', 'koi_insol', 'koi_duration', 'koi_teq', and 'koi_time0bk'. These features were identified as the most significant based on their contributions to the principal components and their importance in the random forest model.

Fig. 2 shows the graph of the Explained Variance Ratio for PCA and the features and their importance using PCA.

C. Univariate Analysis on KOI Data

- **koi_fpflag_ss** (Stellar Eclipse Flag): This feature clearly distinguishes exoplanet dispositions. Candidates and confirmed exoplanets rarely have a stellar eclipse flag, while false positives often do. This indicates that stellar eclipses are a major cause of false positive detections in exoplanet surveys.
- **koi_fpflag_co** (Centroid Offset Flag): Centroid offsets are rare in candidates and confirmed exoplanets but common in false positives, suggesting that these offsets are often linked to false detections due to background stars or instrumental issues.
- **koi_fpflag_ec** (Ephemeris Match Indicates Contamination Flag): Rare in candidates and confirmed exoplanets; common in false positives, suggesting contamination is a key indicator of false positives.
- **koi_fpflag_nt** (Non-Transit-Like Flag): Rare in candidates and confirmed exoplanets; frequent in false positives, indicating non-transit characteristics often signal false detections.

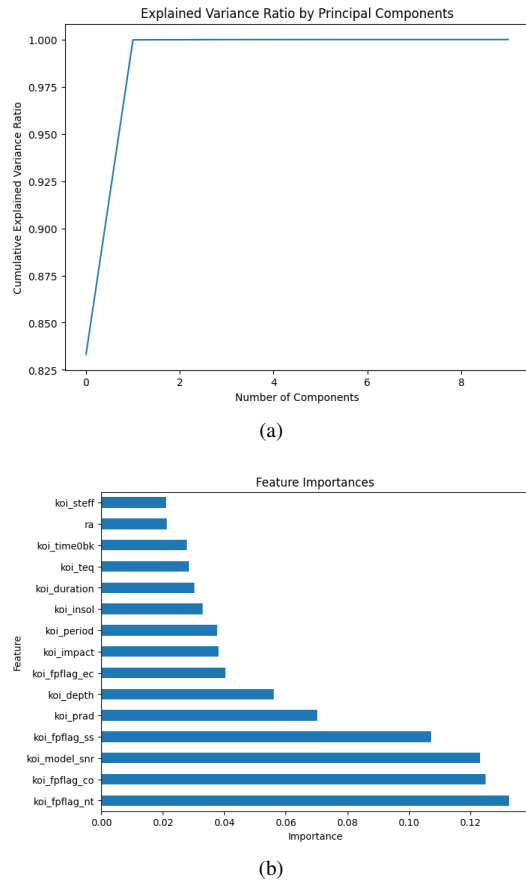


Fig. 2: (a) Explained Variance Ratio, (b) Feature Importance using PCA

- **koi_slogg** (Surface Gravity): Confirmed exoplanets have a narrow range of surface gravity, while false positives have a broader range, reflecting variable host star characteristics.
- **koi_steff** (Stellar Effective Temperature): Confirmed exoplanets are associated with stars of consistent temperatures; false positives show a wider range, indicating diverse star types.
- **koi_srad** (Stellar Radius): Confirmed exoplanets orbit smaller stars with a narrow range of radii; false positives include a broader range, possibly due to larger stars or non-planetary objects.
- **koi_teq** (Equilibrium Temperature): Confirmed exoplanets have stable equilibrium temperatures; false positives and candidates have broader ranges, with false positives extending to extremes.
- **koi_insol** (Insolation Flux): Confirmed exoplanets receive consistent insolation flux; false positives show extreme values, indicating noise or non-planetary phenomena. Candidates are in between.
- **koi_prad** (Planetary Radius): Confirmed exoplanets have realistic planetary radii; false positives include extreme values. Candidates mix valid and false signals.
- **koi_depth** (Transit Depth): Confirmed exoplanets have

consistent transit depths; false positives show a wide range, including shallow and deep transits, suggesting non-planetary phenomena.

- **koi_period** (Orbital Period): Confirmed exoplanets span a wide range of periods; false positives have a narrower range, possibly due to detection biases. Candidates vary widely.
- **koi_time0bk** (Transit Epoch): Confirmed exoplanets have consistent transit timing; false positives show variability, suggesting unreliable timing or artifacts.
- **koi_impact** (Impact Parameter): Confirmed exoplanets generally have lower impact parameters, indicating central transits. False positives have a wider range, including high parameters.
- **koi_duration** (Transit Duration): Confirmed exoplanets show realistic transit durations; false positives often have longer durations, indicating non-planetary events or errors.
- **koi_model_snr** (Transit Signal-to-Noise Ratio): Confirmed exoplanets have higher SNRs, indicating clear detections; false positives have a wide range, often with low SNRs, suggesting poor data quality.

Univariate analysis shows confirmed exoplanets have consistent, narrow ranges across features, indicating stable detections. False positives display broader, more extreme ranges, suggesting measurement errors, noise, or non-planetary phenomena. Candidates fall in between, mixing real and spurious signals.

D. Bivariate Analysis on KOI Data

Analysis of 'dec' and 'ra' values reveals similar ranges across exoplanet classifications (candidates, confirmed, false positives), suggesting they have limited power for distinguishing between them. These coordinates mainly provide positional information rather than insights into exoplanet properties. Bivariate analysis confirms that while 'ra' and 'dec' are essential for positioning, other features related to exoplanet properties, host stars, and transit characteristics are more critical for accurate classification.

E. Machine Learning Models

Table I shows the performance of the machine learning models namely Random Forest, Decision Tree, and AdaBoost on all the 19 features and on selected features. The Random Forest, Decision Tree, and AdaBoost classifiers show that using selected features performs nearly as well as using all features. The accuracy, precision, recall, and F1 score are similar with both approaches, indicating that the chosen features capture the essential data for classification.

F. Probabilistic Models

Table II shows the performance of the probabilistic models namely HMM, Gaussian Processes, Logistic Regression on all the 19 features and on selected features. Comparing the three probabilistic classifiers reveals that while Gaussian Process performs well with all features, Logistic Regression struggles

TABLE I: Performance of the ML models

	Random Forest		Decision Tree		AdaBoost	
Features	All	Selected	All	Selected	All	Selected
Accuracy	0.907	0.897	0.872	0.857	0.883	0.887
Precision	0.907	0.899	0.872	0.857	0.884	0.889
Recall	0.907	0.898	0.872	0.857	0.883	0.887
F1 Score	0.907	0.898	0.872	0.857	0.884	0.886

TABLE II: Performance of the Probabilistic Models

	HMM		Gaussian Processes		Logistic Regression	
Features	All	Selected	All	Selected	All	Selected
Accuracy	0.4718	0.4578	0.7645	0.5324	0.6007	0.5474
Precision	0.5188	0.6017	0.7688	0.5248	0.5798	0.3999
Recall	0.4718	0.4578	0.7645	0.5324	0.6007	0.5474
F1 Score	0.4695	0.4845	0.7640	0.4245	0.5830	0.4592

with fewer features, suggesting important information may be missing. The HMM's consistent performance across feature sets indicates that the selected features may not capture key sequential patterns. This underscores the need for careful feature selection tailored to each model and a deep understanding of the data and modeling methods for optimal performance.

G. Analysis of the Chi Square Test

The analysis yielded the following insights regarding significant associations. Variables found to have statistically significant relationships with the target variable 'koi_disposition' include 'koi_fpflag_nt,' 'koi_fpflag_ss,' 'koi_fpflag_co,' 'koi_fpflag_ec,' 'koi_impact,' 'koi_depth,' 'koi_teq,' 'koi_insol,' 'koi_model_snr,' 'koi_steff,' 'koi_slogg,' 'koi_srad,' 'ra,' 'dec,' and 'koi_kepmag.' These variables exhibit significant associations, indicating their potential importance in predicting exoplanet classifications.

On the other hand, variables such as 'koi_period,' 'koi_time0bk,' 'koi_duration,' and 'koi_prad' did not show significant associations with the target variable. These findings suggest that these variables may have lesser predictive relevance in the context of exoplanet classification.

H. Analysis of the ANOVA Test

The results indicate that variables such as 'koi_fpflag_ss,' 'koi_fpflag_co,' 'koi_fpflag_ec,' 'koi_period,' 'koi_time0bk,' 'koi_impact,' 'koi_duration,' 'koi_depth,' 'koi_teq,' 'koi_insol,' 'koi_model_snr,' 'koi_tce_plnt_num,' 'koi_tce_delivname_encoded,' 'koi_steff,' 'koi_slogg,' 'koi_srad,' 'ra,' 'dec,' and 'koi_kepmag' show significant differences between the groups, suggesting their potential importance in predicting exoplanet classifications. In contrast, variables like 'koi_fpflag_nt' and 'koi_prad' do not show significant differences, implying lesser predictive relevance.

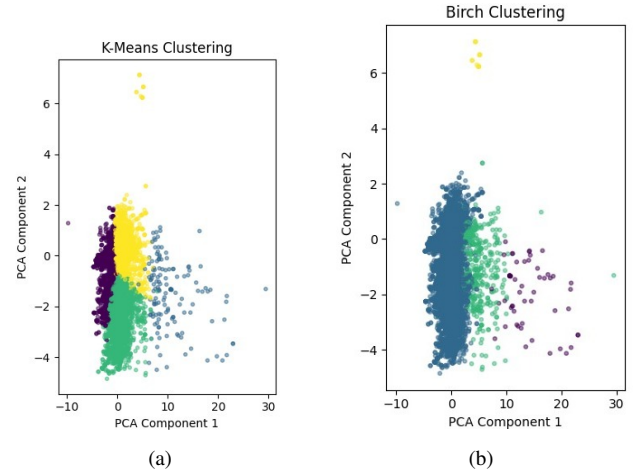


Fig. 3: (a) K-means, (b) BIRCH 2D Representation of the Clustering Methods with PCA

I. Analysis of the Fisher Exact Test

Significant associations were found for variables such as 'koi_score,' 'koi_fpflag_ss,' 'koi_fpflag_co,' 'koi_fpflag_ec,' 'koi_impact,' 'koi_depth,' 'koi_teq,' 'koi_insol,' 'koi_model_snr,' 'koi_tce_plnt_num,' 'koi_tce_delivname_encoded,' 'koi_steff,' 'koi_slogg,' 'koi_srad,' 'ra,' 'dec,' and 'koi_kepmag.' These variables exhibit statistically significant relationships with 'koi_disposition,' highlighting their potential importance in exoplanet classification. Conversely, variables such as 'koi_period,' 'koi_time0bk,' 'koi_duration,' and 'koi_prad' showed no significant associations, suggesting they may be less relevant for this classification task.

J. Clustering on Planetary Systems Data

For K-Means clustering, the Silhouette Score of 0.38 suggests that the clusters are somewhat compact and well-separated, indicating moderate quality clustering. However, the Davies-Bouldin Index of 0.85 indicates a slightly higher degree of clustering overlap and less distinct cluster boundaries.

On the other hand, for BIRCH clustering, the Silhouette Score of 0.60 indicates higher cluster cohesion and better separation compared to K-Means, suggesting more distinct and well-defined clusters. Additionally, the Davies-Bouldin Index of 0.68 suggests lower clustering overlap and clearer boundaries between clusters, indicating better overall clustering quality compared to K-Means. BIRCH clustering outperformed K-Means in cluster quality with PCA dimensionality reduction, as indicated by a higher Silhouette Score and lower Davies-Bouldin Index, showing more cohesive and well-separated clusters.

Fig. 3 and Fig. 4 shows the 2D and 3D image of the K-Means and BIRCH Clustering Methods with PCA.

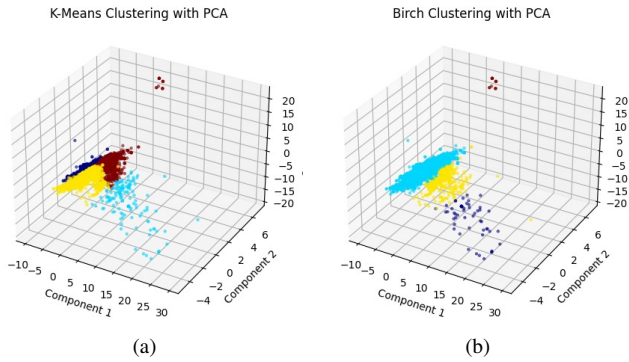


Fig. 4: (a) K-means, (b) BIRCH 3D Representation of the Clustering Methods with PCA

V. DISCUSSIONS

The correlation analysis revealed specific features highly correlated with each other, guiding further analysis towards these informative attributes. This process helped streamline subsequent modeling efforts by focusing on the most relevant variables. Additionally, PCA analysis provided valuable insights into the variance captured by different principal components, enabling dimensionality reduction while preserving essential information. Feature importance analysis validated the significance of selected features, ensuring that only the most informative attributes were retained for modeling.

The univariate and bivariate analyses shed light on the characteristics of exoplanetary systems and their classifications. Notably, certain features exhibited distinct patterns across different classes, facilitating the differentiation between confirmed exoplanets, candidates, and false positives.

Machine learning models used demonstrated robust performance, with selected features yielding comparable results to using all features. This underscores the importance of feature selection in maximizing computational efficiency without sacrificing predictive accuracy. On the other hand, probabilistic models showcased varying performance depending on the feature set used, emphasizing the necessity of precise feature selection tailored to each model's requirements.

The chi-square, ANOVA, and Fisher Exact tests provided valuable insights into the relationships between predictor variables and exoplanet classifications. Significant associations were found for several features, highlighting their predictive relevance, while others showed no significant relationships, suggesting lesser importance in classification tasks. These statistical analyses helped identify key variables crucial for accurate exoplanet classification.

Clustering analysis using K-Means and BIRCH algorithms revealed distinct cluster qualities, with BIRCH outperforming K-Means in terms of cohesion and separation. The visualization of clustering results provided clear insights into the structure of the data, aiding in the interpretation of cluster characteristics and their implications for exoplanetary classification.

Overall, the comprehensive analysis of the data using various statistical techniques and machine learning models yielded valuable insights into the properties and classifications of exoplanetary systems.

VI. CONCLUSION

Analysis of exoplanetary data using statistical techniques and machine learning models has been very useful in understanding their characteristics and classification. Initial correlation analysis identified key attributes that were considered when subsequently modeling with the most informative variables. The PCA indeed simplified dataset complexity adequately retaining relevant information, thus assisting dimensionality reduction. 1D and 2D analyses further unravelled patterns among various exoplanet classes, with an emphasis on concentrating on exoplanetary properties for improved classification accuracy. ML models used performed well with selected features demonstrating the significance of feature selection to enhance computational efficiency without loss in prediction accuracy.

Furthermore, different probabilistic models presented divergent performances thereby underlining the need to select precise features fit for each model's requirements. Statistical tests such as chi-square, ANOVA and Fisher Exact tests also showed which predictor variables are significantly associated with classification of exoplanets thus highlighting important features necessary for classifying accurately. Clustering analysis showcased BIRCH's superiority in terms of cluster cohesion and separation, providing clearer insights into data structure. This study enhances understanding of exoplanetary systems and paves the way for future research by demonstrating that combining statistical techniques with machine learning effectively analyzes and classifies exoplanetary data.

As a result, improved exoplanet classification can be achieved in the future by employing advanced feature selection techniques, investigating deep learning models, incorporating time-series analysis, utilising larger and more diverse datasets, integrating interdisciplinary efforts, and developing real-time classification systems. Such enhancements will improve forecast accuracy, expose detailed patterns, increase model generalisation, and simplify exoplanet detection and confirmation processes, all of which will contribute to a better knowledge of planetary systems.

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